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1 Miscellaneous

1.1 What to Look for

Wrong answer:

Print your solution! (Print debug output, as well.)

Are you clearing all DS between test cases?

Can your algorithm handle the whole range of input?

Read the full problem statement again.

Do you handle all corner cases correctly?

Have you understood the problem correctly?

Any uninitialized variables? Any overflows?

Confusing N and M, i and j, etc.?

Are you sure your algorithm works?

What special cases have you not thought of?

Are you sure the STL functions you use work as you think?

Add some assertions, maybe resubmit.

Create some testcases to run your algorithm on.

Go through the algorithm for a simple case.

Go through this list again.

Go for a small walk, e.g. to the toilet. Is your output format correct? (including ↵ whitespace)

Recode

Runtime error: In some judges, MLE is considered RTE (Check your memory use)

Have you tested all corner cases locally?

Any uninitialized variables?

Are you reading or writing outside the range of any vector?

Any assertions that might fail?

Infinite recursion?

Any possible division by 0? Ex. mod 0

Any possible infinite recursion?

Invalidated pointers or iterators?

Are you using too much memory?

Debug with resubmits.

Force WA (asserts for RTE problems)

Time limit exceeded:

Do you have any possible infinite loops?

What is the complexity of your algorithm?

Are you copying a lot of unnecessary data? (References)

How big is the input and output? (consider scanf)

Avoid vector, map. (use arrays/unordered_map)

What do your teammates think about your algorithm?

Memory limit exceeded:

What is the max amount of memory your algorithm should need?

Are you clearing all DS between test cases?

Infinite recursion? (As in pushing into a DS infinitely)

1.2 Day of Date

```
// 0-based
const vector<int> T = {0, 3, 2, 5, 0, 3, 5, 1, 4, 6, 2, 4}
int day(int d, int m, int y) {
    y -= (m < 3);
    return (y + y / 4 - y / 100 + y / 400 + T[m - 1] + d) % 7;
}
```

1.3 Number of Days since 1-1-1

```
int rdn(int d, int m, int y) {
    if(m < 3)
        --y, m += 12;
    return 365 * y + y / 4 - y / 100 + y / 400
        + (153 * m - 457) / 5 + d - 306;
}
```

1.4 Enumerate Subsets of a Bitmask

```
int x = 0;
do {
    // do stuff with the bitmask here
    x = (x + 1 + ~m) & m;
} while(x != 0);
```

1.5 Josephus Problem

```
ll josephus(ll n, ll k) { // O(k log n)
    if(n == 1) return 0;
    if(k == 1) return n - 1;
    if(k > n) return (josephus(n - 1, k) + k) % n;
    ll cnt = n / k;
    ll res = josephus(n - cnt, k);
    res -= n % k;
    if(res < 0) res += n;
    else res += res / (k - 1);
    return res;
}
int josephus(int n, int k) { // O(n)
    int res = 0;
    for(int i = 1; i <= n; ++i) res = (res + k) % i;
    return res + 1;
}
```

1.6 Random Primes

```
36671 74101 724729 825827 924997 1500005681 2010408371 2010405347
```

1.7 RNG

```
mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());

// [mn, mx]
template<class T> T rand_int(T mn, T mx) {
    return uniform_int_distribution<T>(mn, mx)(rng);
}
// [mn,mx) NOT TYPO
template<class T> T rand_float(T mn, T mx) {
    return uniform_real_distribution<T>(mn, mx)(rng);
}
```

2 Data Structures

2.1 Segment Tree

```
struct segment_tree {
    ll SZ=1;
    vector<ll> arr;
    ll default_value = 0;
    segment_tree(int _sz=(int)2.5e5) {
        while (SZ<_sz) {SZ<<=1;}
        arr.resize(SZ<<1);
        for (auto& x:arr) x=default_value;
    }

    ll merge(ll a, ll b) {
        return a+b;
    }

    void update(ll idx, ll val) {
        ll nd=idx+SZ;
        arr[nd]=val;
        while (nd>1) { //update par
            nd/=2;
            arr[nd] = merge(arr[2*nd], arr[2*nd+1]);
        }
    }

    ll query(ll nd, ll cl, ll cr, ll ql, ll qr) {
        if (ql<=cl && cr<=qr) return arr[nd];
        else if (cl>qr || cr<ql) return default_value;
        else {
```

```
            ll mid = (cl+cr)/2;
            return merge(query(2*nd, cl, mid, ql, qr), query(2*nd+1, mid+1, cr, ql, qr));
        }
    }
};
```

2.2 Fenwick RU-RQ

```
void updtRL(int l, int r, ll val) {
    updt(BIT1, l, val), updt(BIT1, r + 1, -val);
    updt(BIT2, l, val * (l - 1)), updt(BIT2, r + 1, -val * r);
}
ll query(int k) {
    return que(BIT1, k) * k - que(BIT2, k);
}
```

2.3 Heavy-Light Decomposition

```
struct HLD {
    int n;
    vector<int> id, size, idx, up, root, st;
    vector<vector<int>> adj, chain;
    SegTree seg;

    HLD(const vector<vector<int>>& edges) :
        n(edges.size()), id(n, -1), size(n, -1), idx(n, -1),
        up(n, -1), adj(edges), seg(n) {
        precompute(0, -1);
        decompose(0, -1);
        int cnt = 0;
        st.resize(chain.size());
        for(int i = 0; i < (int) chain.size(); ++i) {
            st[i] = cnt;
            cnt += chain[i].size();
        }
    }

    void precompute(int pos, int dad) {
        size[pos] = 1;
        up[pos] = dad;
        for(auto& i : adj[pos]) {
            if(i != dad) {
                precompute(i, pos);
                size[pos] += size[i];
            }
        }
    }

    void decompose(int pos, int dad) {
        if(id[pos] == -1) {
            id[pos] = chain.size();
            root.push_back(pos);
            chain.emplace_back();
        }
        idx[pos] = chain[id[pos]].size();
        chain[id[pos]].push_back(pos);
        int mx = 0, heavy = -1;
        for(auto& i : adj[pos]) {
            if(i != dad && size[i] > mx) {
                mx = size[i];
                heavy = i;
            }
        }
        if(heavy != -1)
            id[heavy] = id[pos];
        for(auto& i : adj[pos]) {
            if(i != dad)
                decompose(i, pos);
        }
    }
};
```

```

    }
}

void update(int ch, int l, int r, int val) {
    seg.update(st[ch] + l, st[ch] + r, val);
}

int query(int ch, int l, int r, int val) {
    return seg.query(st[ch] + l, st[ch] + r, val);
}
};

// how to move from u to v
while(1) {
    if(hld.id[u] == hld.id[v]) {
        if(hld.idx[u] > hld.idx[v])
            swap(u, v);
        hld.update(hld.id[u], hld.idx[u], hld.idx[v], w);
        // or hld.query(hld.id[u], hld.idx[u], hld.idx[v]);
        break;
    }
    if(hld.id[u] < hld.id[v])
        swap(u, v);
    hld.update(hld.id[u], 0, hld.idx[u], w);
    // or hld.query(hld.id[u], 0, hld.idx[u]);
    u = hld.up[hld.root[hld.id[u]]];
}

```

2.4 Li-Chao Tree

```

// max li-chao tree
// works for the range [0, MAX - 1]
// if min li-chao tree:
// replace every call to max() with min() and every > with <
// also replace -INF with INF

struct Func {
    ll m, c;
    ll operator()(ll x) {
        return x * m + c;
    }
};

const int MAX = 1e9 + 1;
const ll INF = 1e18;
const Func NIL = {0, -INF};

struct Node {
    Func f;
    Node* lc;
    Node* rc;

    Node() : f(NIL), lc(nullptr), rc(nullptr) {}
    Node(const Node& n) : f(n.f), lc(nullptr), rc(nullptr) {}
};

Node* root = new Node;

void insert(Func f, Node* cur = root, int l = 0, int r = MAX - 1) {
    int m = l + (r - l) / 2;
    bool left = f(l) > cur->f(l);
    bool mid = f(m) > cur->f(m);
    if(mid)
        swap(f, cur->f);
    if(l != r) {
        if(left != mid) {
            if(!cur->lc)
                cur->lc = new Node(*cur);
            insert(f, cur->lc, l, m);
        } else {

```

```

            if(!cur->rc)
                cur->rc = new Node(*cur);
            insert(f, cur->rc, m + 1, r);
        }
    }
}

ll query(ll x, Node* cur = root, int l = 0, int r = MAX - 1) {
    if(!cur)
        return -INF;
    if(l == r)
        return cur->f(x);
    int m = l + (r - l) / 2;
    if(x <= m)
        return max(cur->f(x), query(x, cur->lc, l, m));
    else
        return max(cur->f(x), query(x, cur->rc, m + 1, r));
}

```

2.5 STL PBDS

```

// ost = ordered set
// omp = ordered map
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>

using namespace __gnu_pbds;

template<class T>
using ost = tree<T, null_type, less<T>, rb_tree_tag,
tree_order_statistics_node_update>;
template<class T, class U>
using omp = tree<T, U, less<T>, rb_tree_tag,
tree_order_statistics_node_update>;

```

2.6 Treap

```

// Complexity: O(log N) for split and merge
//
// empty treap: Treap* tr = nullptr;
// insert v at x: [l, r] = split(tr, x), m = Treap(v), merge lmr
// delete at x: [l, r] = split(tr, x), [m, r] = split(r, l), merge lr
// lazy prop: propagate every time a node is accessed

mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());

using Key = int;

struct Treap {
    Key val;
    Treap* left;
    Treap* right;
    int prio, sz;
    Treap() {}
    Treap(int _val);
};

int size(Treap* tr) {
    return tr ? tr->sz : 0;
}

void update(Treap* tr) {
    tr->sz = 1 + size(tr->left) + size(tr->right);
}

Treap::Treap(Key _val) :
    val(_val), left(nullptr), right(nullptr), prio(rng()) {
    update(this);
}

```

```

pair<Treap*, Treap*> split(Treap* tr, int sz) {
    if(!tr) return {nullptr, nullptr};
    int left_sz = size(tr->left);
    if(sz <= left_sz) {
        auto [left, mid] = split(tr->left, sz);
        tr->left = mid;
        update(tr);
        return {left, tr};
    } else {
        auto [mid, right] = split(tr->right, sz - left_sz - 1);
        tr->right = mid;
        update(tr);
        return {tr, right};
    }
}

Treap* merge(Treap* l, Treap* r) {
    if(!l) return r;
    if(!r) return l;
    if(l->prio < r->prio) {
        l->right = merge(l->right, r);
        update(l);
        return l;
    } else {
        r->left = merge(l, r->left);
        update(r);
        return r;
    }
}

```

2.7 Unordered Map Custom Hash

```

struct custom_hash {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }
    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM =
            chrono::steady_clock::now().time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};

unordered_map<int, int, custom_hash> umap;

```

2.8 Mo's on Tree

$ST(u) \leq ST(v)$
 $P = LCA(u, v)$
 If $P = u$, query $[ST(u), ST(v)]$
 Else query $[EN(u), ST(v)] + [ST(P), ST(P)]$

2.9 Link-Cut Tree

```

// sz for path queries
// sub, vsub for subtree queries
// x->access() brings x to the top and propagates it,
// x's left subtree is path from x to root
// after access, sub is no. of nodes in CC of x,
// vsub is no. of nodes under x
// use makeRoot() for arbitrary path queries

```

```

typedef struct Node* Np;
struct Node {
    Np p, c[2]; // parent, children
    bool flip = 0; // subtree flipped or not
    int vtx;
    ll val, sz; // value in node, # nodes in current splay tree
    int sub, vsub = 0; // # of nodes in CC
    ll stsum; // sum of all val in the splay tree
    // after access(): from original root to this node
    Node(int vtx, int _val = 1) : vtx(_vtx), val(_val) {
        p = c[0] = c[1] = nullptr;
        calc();
    }

    friend int getSz(Np x) {
        return x ? x->sz : 0;
    }
    friend int getSub(Np x) {
        return x ? x->sub : 0;
    }
    friend ll getStSum(Np x) {
        return x ? x->stsum : 0;
    }

    void prop() { // lazy prop
        if(!flip) return;
        swap(c[0], c[1]);
        flip = 0;
        for(int i = 0; i < 2; i++) {
            if(c[i]) c[i]->flip ^= 1;
        }
    }
    void calc() { // recalc vals
        for(int i = 0; i < 2; i++) {
            if(c[i]) c[i]->prop();
        }
        sz = 1 + getSz(c[0]) + getSz(c[1]);
        sub = 1 + getSub(c[0]) + getSub(c[1]) + vsub;
        stsum = val + getStSum(c[0]) + getStSum(c[1]);
    }

    // SPLAY TREE OPERATIONS
    int dir() { // p is path-parent pointer
        if(!p) return -2;
        for(int i = 0; i < 2; i++) {
            if(p->c[i] == this) return i;
        }
        return -1; // -> not in current splay tree
    }
    // test if root of current splay tree
    bool isRoot() {
        return dir() < 0;
    }
    friend void setLink(Np x, Np y, int d) { // x is orig parent
        if(y) y->p = x;
        if(d >= 0) x->c[d] = y;
    }
    void rot() { // assume p and p->p propagated
        assert(!isRoot());
        int x = dir();
        Np pa = p;
        setLink(pa->p, this, pa->dir());
        setLink(pa, c[x ^ 1], x);
        setLink(this, pa, x ^ 1);
    }
}

```

```

    pa->calc();
}
void splay() { // bring this node to the root of splay tree
    while(!isRoot() && !p->isRoot()) {
        p->p->prop(), p->prop(), prop();
        dir() == p->dir() ? p->rot() : rot();
        rot();
    }
    if(!isRoot())
        p->prop(), prop(), rot();
    prop();
    calc();
}
Np fbo(int b) { // find by order
    prop();
    int z = getSz(c[0]); // of splay tree
    if(b == z) {
        splay();
        return this;
    }
    return b < z ? c[0]->fbo(b) : c[1]->fbo(b - z - 1);
}

// BASE OPERATIONS
// bring this to top of splay tree (not impacting original tree)
void access() {
    for(Np v = this, pre = nullptr; v; v = v->p) {
        v->splay(); // now switch virtual children
        if(pre)
            v->vsub -= pre->sub;
        if(v->c[1])
            v->vsub += v->c[1]->sub;
        v->c[1] = pre;
        v->calc();
        pre = v;
    }
    splay();
    assert(!c[1]); // right subtree is empty
}
void makeRoot() { // of the original tree
    access();
    flip ^= 1;
    access();
    assert(!c[0] && !c[1]);
}

// QUERIES
friend Np lca(Np x, Np y) {
    if(x == y)
        return x;
    x->access(), y->access();
    if(!x->p)
        return nullptr;
    x->splay();
    return x->p ? x : x; // y was below x in latter case
} // access at y did not affect x -> not connected
friend bool connected(Np x, Np y) {
    return lca(x, y);
}

// no. of nodes above; distance to root in original tree
int distRoot() {
    access();
    return getSz(c[0]);
}
Np getRoot() { // get root of LCT component in original tree
    access();
    Np a = this;
    while(a->c[0])
        a = a->c[0], a->prop();
    a->access();

```

```

    return a;
}
Np getPar(int b) { // get b-th parent on path to root
    access();
    b = getSz(c[0]) - b;
    assert(b >= 0);
    return fbo(b);
} // can also get min, max on path to root, etc

// MODIFICATIONS
void setVal(int v) {
    access();
    val = v;
    calc();
}
void addVal(int v) {
    access();
    val += v;
    calc();
}
}
friend void link(Np x, Np y, bool force = 1) {
    assert(!connected(x, y));
    if(force) {
        y->makeRoot(); /// make x par of y; x -> y
    } else {
        y->access();
        assert(!y->c[0]);
    }
    x->access();
    setLink(y, x, 0);
    y->calc();
}
friend void cut(Np y) { // cut y from its parent
    y->access();
    assert(y->c[0]);
    y->c[0]->p = NULL;
    y->c[0] = NULL;
    y->calc();
}
friend void cut(Np x, Np y) { // if x, y adj in tree
    x->makeRoot();
    y->access();
    assert(y->c[0] == x && !x->c[0] && !x->c[1]);
    cut(y);
}
};
Np LCT[maxV];

```

2.10 Mo's on Array

```

void remove(idx); // TODO: remove value at idx from data structure
void add(idx); // TODO: add value at idx from data structure
int get_answer(); // TODO: extract the current answer of the data structure

int block_size;

struct Query {
    int l, r, idx;
    bool operator<(Query other) const
    {
        return make_pair(l / block_size, r) <
            make_pair(other.l / block_size, other.r);
    }
};

vector<int> mo_s_algorithm(vector<Query> queries) {
    vector<int> answers(queries.size());
    sort(queries.begin(), queries.end());

    // TODO: initialize data structure

```

```

int cur_l = 0;
int cur_r = -1;
// invariant: data structure will always reflect the range [cur_l, cur_r]
for (Query q : queries) {
    while (cur_l > q.l) {
        cur_l--;
        add(cur_l);
    }
    while (cur_r < q.r) {
        cur_r++;
        add(cur_r);
    }
    while (cur_l < q.l) {
        remove(cur_l);
        cur_l++;
    }
    while (cur_r > q.r) {
        remove(cur_r);
        cur_r--;
    }
    answers[q.idx] = get_answer();
}
return answers;
}

```

3 Dynamic Programming

3.1 DP CHT

```

#include<complex> // Required!!!
typedef int ftype; // ll/ld/int will do
typedef complex<ftype> point;
#define x real
#define y imag

ftype dot(point a, point b) { return (conj(a) * b).x(); }
ftype cross(point a, point b) { return (conj(a) * b).y(); }

struct CHT {
    // Currently min, to switch to max, flip all <0 and >0.
    vector<point> hull, vecs;

    void add_line(ftype k, ftype b) {
        point nw = {k, b};
        while (!vecs.empty() && dot(vecs.back(), nw - hull.back()) < 0) {
            hull.pop_back();
            vecs.pop_back();
        }
        if (!hull.empty()) vecs.push_back(point(0, 1) * (nw - hull.back()));
        hull.push_back(nw);
    }

    ftype get(ftype x) const {
        point query = {x, 1};
        auto it = lower_bound(vecs.begin(), vecs.end(), query,
                               [](point a, point b) { return cross(a, b) > 0; });
        int idx = it - vecs.begin();
        if (idx == (int)hull.size()) idx--;
        return dot(query, hull[idx]);
    }
};

```

3.2 DP DNC

```

void f(int rem, int l, int r, int optl, int optr) {
    if (l > r)
        return;
    int mid = l + r >> 1;

```

```

int opt = MOD, optid = mid;
for (int i = optl; i <= mid && i <= optr; ++i) {
    if (dp[rem - 1][i] + c[i][mid] < opt) {
        opt = dp[rem - 1][i] + c[i][mid];
        optid = i;
    }
}
dp[rem][mid] = opt;
f(rem, l, mid - 1, optl, optid);
f(rem, mid + 1, r, optid, optr);
return;
}
rep(i, 1, n) dp[1][i] = c[0][i];
rep(i, 2, k) f(i, i, n, i, n);

```

3.3 DP Knuth-Yao

```

// opt[i+1][j] <= opt[i][j] <= opt[i][j+1]
// dp[i][j] = min{k} dp[i][k] + dp[k][j] + cost[i][j]
for (int k = 0; k <= n; k++) {
    for (int i = 0; i + k <= n; i++) {
        if (k < 2)
            dp[i][i + k] = 0, opt[i][i + k] = i;
        else {
            int sta = opt[i][i + k - 1];
            int end = opt[i + 1][i + k];
            for (int j = sta; j <= end; j++) {
                if (dp[i][j] + dp[j][i + k] + cost[i][i + k] < dp[i][i + k]) {
                    dp[i][i + k] = dp[i][j] + dp[j][i + k] + cost[i][i + k];
                    opt[i][i + k] = j;
                }
            }
        }
    }
}
}
}

```

3.4 DP SOS

```

vector<int> sos(1 << n);
vector<vector<int>> dp(1 << n, vector<int>(n + 1));

for (int x = 0; x < (1 << n); x++) {
    dp[x][0] = a[x];
    for (int i = 0; i < n; i++) {
        dp[x][i + 1] = dp[x][i];
        if (x & (1 << i)) { dp[x][i + 1] += dp[x ^ (1 << i)][i]; }
    }
    sos[x] = dp[x][n];
}

// Alternative memory optimized
sos = a;

for (int i = 0; i < n; i++) {
    for (int x = 0; x < (1 << n); x++) {
        if (x & (1 << i)) { sos[x] += sos[x ^ (1 << i)]; }
    }
}

```

4 Geometry

4.1 Geometry Template

```

/*
TABLE OF CONTENT
0. Basic Rule
    0.1. Everything is in double

```

```

0.2. Every comparison use EPS
0.3. Every degree in rad
1. General Double Operation
1.1. const double EPS=1E-9
1.2. const double PI=acos(-1.0)
1.3. const double INFD=1E9
1.3. between_d(double x,double l,double r)
    check whether x is between l and r inclusive with EPS
1.4. same_d(double x,double y)
    check whether x=y with EPS
1.5. dabs(double x)
    absolute value of x
2. Point
2.1. struct point
2.1.1. double x,y
    cartesian coordinate of the point
2.1.2. point()
    default constructor
2.1.3. point(double _x,double _y)
    constructor, set the point to (_x,_y)
2.1.4. bool operator< (point other)
    regular pair<double,double> operator < with EPS
2.1.5. bool operator== (point other)
    regular pair<double,double> operator == with EPS
2.2. hypot(point P)
    length of hypotenuse of point P to (0,0)
2.3. e_dist(point P1,point P2)
    euclidean distance from P1 to P2
2.4. m_dist(point P1,point P2)
    manhattan distance from P1 to P2
2.5. point rotate(point P,point O,double angle)
    rotate point P from the origin O by angle ccw
3. Vector
3.1. struct vec
3.1.1. double x,y
    x and y magnitude of the vector
3.1.2. vec()
    default constructor
3.1.3. vec(double _x,double _y)
    constructor, set the vector to (_x,_y)
3.1.4. vec(point A,point B)
    constructor, set the vector to vector AB (A->B)
*/
/*General Double Operation*/

const double PI = acos(-1.0);
const double INFD = 1E9;
double between_d(double x, double l, double r) {
    return (min(l, r) <= x + EPS && x <= max(l, r) + EPS);
}
double same_d(double x, double y) {
    return between_d(x, y, y);
}
double dabs(double x) {
    if(x < EPS)
        return -x;
    return x;
}
/*Point*/
struct point {
    double x, y;
    point() {
        x = y = 0.0;
    }
    point(double _x, double _y) {
        x = _x;
        y = _y;
    }
    bool operator< (point other) {
        if(x < other.x + EPS)
            return true;

```

```

        if(x + EPS > other.x)
            return false;
        return y < other.y + EPS;
    }
    bool operator== (point other) {
        return same_d(x, other.x) && same_d(y, other.y);
    }
};
double e_dist(point P1, point P2) {
    return hypot(P1.x - P2.x, P1.y - P2.y);
}
double m_dist(point P1, point P2) {
    return dabs(P1.x - P2.x) + dabs(P1.y - P2.y);
}
double pointBetween(point P, point L, point R) {
    return (e_dist(L, P) + e_dist(P, R) == e_dist(L, R));
}
bool collinear(point P, point L,
               point R) { //newly added(luis), cek 3 poin segaris
    return P.x * (L.y - R.y) + L.x * (R.y - P.y) + R.x * (P.y - L.y) ==
           0; // bole gnti "dabs(x)<"EPS
}

/*Vector*/
struct vec {
    double x, y;
    vec() {
        x = y = 0.0;
    }
    vec(double _x, double _y) {
        x = _x;
        y = _y;
    }
    vec(point A) {
        x = A.x;
        y = A.y;
    }
    vec(point A, point B) {
        x = B.x - A.x;
        y = B.y - A.y;
    }
};
vec scale(vec v, double s) {
    return vec(v.x * s, v.y * s);
}
vec flip(vec v) {
    return vec(-v.x, -v.y);
}
double dot(vec u, vec v) {
    return (u.x * v.x + u.y * v.y);
}
double cross(vec u, vec v) {
    return (u.x * v.y - u.y * v.x);
}
double norm_sq(vec v) {
    return (v.x * v.x + v.y * v.y);
}
point translate(point P, vec v) {
    return point(P.x + v.x, P.y + v.y);
}
point rotate(point P, point O, double angle) {
    vec v(0);
    P = translate(P, flip(v));
    return translate(point(P.x * cos(angle) - P.y * sin(angle),
                          P.x * sin(angle) + P.y * cos(angle)), v);
}
point mid(point P, point Q) {
    return point((P.x + Q.x) / 2, (P.y + Q.y) / 2);
}
double angle(point A, point O, point B) {
    vec OA(O, A), OB(O, B);

```

```

    return acos(dot(OA, OB) / sqrt(norm_sq(OA) * norm_sq(OB)));
}
int orientation(point P, point Q, point R) {
    vec PQ(P, Q), PR(P, R);
    double c = cross(PQ, PR);
    if(c < -EPS)
        return -1;
    if(c > EPS)
        return 1;
    return 0;
}
/*Line*/
struct line {
    double a, b, c;
    line() {
        a = b = c = 0.0;
    }
    line(double _a, double _b, double _c) {
        a = _a;
        b = _b;
        c = _c;
    }
    line(point P1, point P2) {
        if(P1 < P2)
            swap(P1, P2);
        if(same_d(P1.x, P2.x))
            a = 1.0, b = 0.0, c = -P1.x;
        else
            a = -(P1.y - P2.y) / (P1.x - P2.x), b = 1.0, c = -(a * P1.x) - P1.y;
    }
    line(point P, double slope) {
        if(same_d(slope, INFD))
            a = 1.0, b = 0.0, c = -P.x;
        else
            a = -slope, b = 1.0, c = -(a * P.x) - P.y;
    }
    bool operator==(line other) {
        return same_d(a, other.a) && same_d(b, other.b) && same_d(c, other.c);
    }
    double slope() {
        if(same_d(b, 0.0))
            return INFD;
        return -(a / b);
    }
};
bool paralel(line L1, line L2) {
    return same_d(L1.a, L2.a) && same_d(L1.b, L2.b);
}
bool intersection(line L1, line L2, point& P) {
    if(paralel(L1, L2))
        return false;
    P.x = (L2.b * L1.c - L1.b * L2.c) / (L2.a * L1.b - L1.a * L2.b);
    if(same_d(L1.b, 0.0))
        P.y = -(L2.a * P.x + L2.c);
    else
        P.y = -(L1.a * P.x + L1.c);
    return true;
}
double pointToLine(point P, point A, point B, point& C) {
    vec AP(A, P), AB(A, B);
    double u = dot(AP, AB) / norm_sq(AB);
    C = translate(A, scale(AB, u));
    return e_dist(P, C);
}
double lineToLine(line L1, line L2) {
    if(!paralel(L1, L2))
        return 0.0;
    return dabs(L2.c - L1.c) / sqrt(L1.a * L1.a + L1.b * L1.b);
}
/*Line Segment*/
struct segment {

```

```

    point P, Q;
    line L;
    segment() {
        point T1;
        P = Q = T1;
        line T2;
        L = T2;
    }
    segment(point _P, point _Q) {
        P = _P;
        Q = _Q;
        if(Q < P)
            swap(P, Q);
        line T(P, Q);
        L = T;
    }
    bool operator==(segment other) {
        return P == other.P && Q == other.Q;
    }
};
bool onSegment(point P, segment S) {
    if(orientation(S.P, S.Q, P) != 0)
        return false;
    return between_d(P.x, S.P.x, S.Q.x) && between_d(P.y, S.P.y, S.Q.y);
}
bool s_intersection(segment S1, segment S2) {
    double o1 = orientation(S1.P, S1.Q, S2.P);
    double o2 = orientation(S1.P, S1.Q, S2.Q);
    double o3 = orientation(S2.P, S2.Q, S1.P);
    double o4 = orientation(S2.P, S2.Q, S1.Q);
    if(o1 != o2 && o3 != o4)
        return true;
    if(o1 == 0 && onSegment(S2.P, S1))
        return true;
    if(o2 == 0 && onSegment(S2.Q, S1))
        return true;
    if(o3 == 0 && onSegment(S1.P, S2))
        return true;
    if(o4 == 0 && onSegment(S1.Q, S2))
        return true;
    return false;
}
double pointToSegment(point P, point A, point B, point& C) {
    vec AP(A, P), AB(A, B);
    double u = dot(AP, AB) / norm_sq(AB);
    if(u < EPS) {
        C = A;
        return e_dist(P, A);
    }
    if(u + EPS > 1.0) {
        C = B;
        return e_dist(P, B);
    }
    return pointToLine(P, A, B, C);
}
double segmentToSegment(segment S1, segment S2) {
    if(s_intersection(S1, S2))
        return 0.0;
    double ret = INFD;
    point dummy;
    ret = min(ret, pointToSegment(S1.P, S2.P, S2.Q, dummy));
    ret = min(ret, pointToSegment(S1.Q, S2.P, S2.Q, dummy));
    ret = min(ret, pointToSegment(S2.P, S1.P, S1.Q, dummy));
    ret = min(ret, pointToSegment(S2.Q, S1.P, S1.Q, dummy));
    return ret;
}
/*Circle*/
struct circle {
    point P;
    double r;
    circle() {

```



```

    point P1;
    P = P1;
    r = 0.0;
}
circle(point _P, double _r) {
    P = _P;
    r = _r;
}
circle(point P1, point P2) {
    P = mid(P1, P2);
    r = e_dist(P, P1);
}
circle(point P1, point P2, point P3) {
    vector<point> T;
    T.clear();
    T.pb(P1);
    T.pb(P2);
    T.pb(P3);
    sort(T.begin(), T.end());
    P1 = T[0];
    P2 = T[1];
    P3 = T[2];
    point M1, M2;
    M1 = mid(P1, P2);
    M2 = mid(P2, P3);
    point Q2, Q3;
    Q2 = rotate(P2, P1, PI / 2);
    Q3 = rotate(P3, P2, PI / 2);
    vec P1Q2(P1, Q2), P2Q3(P2, Q3);
    point M3, M4;
    M3 = translate(M1, P1Q2);
    M4 = translate(M2, P2Q3);
    line L1(M1, M3), L2(M2, M4);
    intersection(L1, L2, P);
    r = e_dist(P, P1);
}
bool operator==(circle other) {
    return (P == other.P && same_d(r, other.r));
}
};
bool insideCircle(point P, circle C) {
    return e_dist(P, C.P) <= C.r + EPS;
}
bool c_intersection(circle C1, circle C2, point& P1, point& P2) {
    double d = e_dist(C1.P, C2.P);
    if(d > C1.r + C2.r) {
        return false; //d+EPS kalo butuh
    }
    if(d < dabs(C1.r - C2.r) + EPS)
        return false;
    double x1 = C1.P.x, y1 = C1.P.y, r1 = C1.r, x2 = C2.P.x, y2 = C2.P.y, r2 = C2.r;
    double a = (r1 * r1 - r2 * r2 + d * d) / (2 * d), h = sqrt(r1 * r1 - a * a);
    point T(x1 + a * (x2 - x1) / d, y1 + a * (y2 - y1) / d);
    P1 = point(T.x - h * (y2 - y1) / d, T.y + h * (x2 - x1) / d);
    P2 = point(T.x + h * (y2 - y1) / d, T.y - h * (x2 - x1) / d);
    return true;
}
bool lc_intersection(line L, circle O, point& P1, point& P2) {
    double a = L.a, b = L.b, c = L.c, x = O.P.x, y = O.P.y, r = O.r;
    double A = a * a + b * b, B = 2 * a * b * y - 2 * a * c - 2 * b * b * x,
        C = b * b * x * x + b * b * y * y - 2 * b * c * y + c * c - b * b * r * r;
    double D = B * B - 4 * A * C;
    point T1, T2;
    if(same_d(b, 0.0)) {
        T1.x = c / a;
        if(dabs(x - T1.x) + EPS > r)
            return false;
        if(same_d(T1.x - r - x, 0.0) || same_d(T1.x + r - x, 0.0)) {
            P1 = P2 = point(T1.x, y);
            return true;
        }
    }
}

```

```

    double dx = dabs(T1.x - x), dy = sqrt(r * r - dx * dx);
    P1 = point(T1.x, y - dy);
    P2 = point(T1.x, y + dy);
    return true;
}
if(same_d(D, 0.0)) {
    T1.x = -B / (2 * A);
    T1.y = (c - a * T1.x) / b;
    P1 = P2 = T1;
    return true;
}
if(D < EPS)
    return false;
D = sqrt(D);
T1.x = (-B - D) / (2 * A);
T1.y = (c - a * T1.x) / b;
P1 = T1;
T2.x = (-B + D) / (2 * A);
T2.y = (c - a * T2.x) / b;
P2 = T2;
return true;
}
bool sc_intersection(segment S, circle C, point& P1, point& P2) {
    bool cek = lc_intersection(S.L, C, P1, P2);
    if(!cek)
        return false;
    double x1 = S.P.x, y1 = S.P.y, x2 = S.Q.x, y2 = S.Q.y;
    bool b1 = between_d(P1.x, x1, x2) && between_d(P1.y, y1, y2);
    bool b2 = between_d(P2.x, x1, x2) && between_d(P2.y, y1, y2);
    if(P1 == P2)
        return b1;
    if(b1 || b2) {
        if(!b1)
            P1 = P2;
        if(!b2)
            P2 = P1;
        return true;
    }
    return false;
}
/*Triangle*/
double t_perimeter(point A, point B, point C) {
    return e_dist(A, B) + e_dist(B, C) + e_dist(C, A);
}
double t_area(point A, point B, point C) {
    double s = t_perimeter(A, B, C) / 2;
    double ab = e_dist(A, B), bc = e_dist(B, C), ac = e_dist(C, A);
    return sqrt(s * (s - ab) * (s - bc) * (s - ac));
}
circle t_inCircle(point A, point B, point C) {
    vector<point> T;
    T.clear();
    T.pb(A);
    T.pb(B);
    T.pb(C);
    sort(T.begin(), T.end());
    A = T[0];
    B = T[1];
    C = T[2];
    double r = t_area(A, B, C) / (t_perimeter(A, B, C) / 2);
    double ratio = e_dist(A, B) / e_dist(A, C);
    vec BC(B, C);
    BC = scale(BC, ratio / (1 + ratio));
    point P;
    P = translate(B, BC);
    line AP1(A, P);
    ratio = e_dist(B, A) / e_dist(B, C);
    vec AC(A, C);
    AC = scale(AC, ratio / (1 + ratio));
    P = translate(A, AC);
    line BP2(B, P);
}

```

```

    intersection(AP1, BP2, P);
    return circle(P, r);
}
circle t_outCircle(point A, point B, point C) {
    return circle(A, B, C);
}
/*Polygon*/
struct polygon {
    vector<point> P;
    polygon() {
        P.clear();
    }
    polygon(vector<point>& _P) {
        P = _P;
    }
};
bool rayCast(point P, polygon& A) {
    point Q(P.x, 10000);
    line cast(P, Q);
    int cnt = 0;
    FOR(i, (int)(A.P.size()) - 1) {
        line temp(A.P[i], A.P[i + 1]);
        point I;
        bool B = intersection(cast, temp, I);
        if(!B)
            continue;
        else if(I == A.P[i] || I == A.P[i + 1])
            continue;
        else if(pointBetween(I, A.P[i], A.P[i + 1]) && pointBetween(I, P, Q))
            cnt++;
    }
    return cnt % 2 == 1;
}
// line segment p-q intersect with line A-B.
point lineIntersectSeg(point p, point q, point A, point B) {
    double a = B.y - A.y;
    double b = A.x - B.x;
    double c = B.x * A.y - A.x * B.y;
    double u = fabs(a * p.x + b * p.y + c);
    double v = fabs(a * q.x + b * q.y + c);
    return point((p.x * v + q.x * u) / (u + v), (p.y * v + q.y * u) / (u + v));
}
// cuts polygon Q along the line formed by point a -> point b
// (note: the last point must be the same as the first point)
vector<point> cutPolygon(point a, point b, const vector<point>& Q) {
    vector<point> P;
    for(int i = 0; i < (int)Q.size(); i++) {
        double left1 = cross(toVec(a, b), toVec(a, Q[i]));
        double left2 = 0;
        if(i != (int)Q.size() - 1)
            left2 = cross(toVec(a, b), toVec(a, Q[i + 1]));
        if(left1 > -EPS)
            P.push_back(Q[i]);
        if(left1 * left2 < -EPS)
            P.push_back(lineIntersectSeg(Q[i], Q[i + 1], a, b));
    }
    if(!P.empty() && !(P.back() == P.front()))
        P.push_back(P.front());
    return P;
}
circle minCoverCircle(polygon& A) {
    vector<point> p = A.P;
    point c;
    circle ret;
    double cr = 0.0;
    int i, j, k;
    c = p[0];
    for(i = 1; i < p.size(); i++) {
        if(e_dist(p[i], c) >= cr + EPS) {
            c = p[i]; cr = 0;
            ret = circle(c, cr);
        }
    }
    for(j = 0; j < i; j++) {
        if(e_dist(p[j], c) >= cr + EPS) {
            c = mid(p[i], p[j]);
            cr = e_dist(p[i], c);
            ret = circle(c, cr);
            for(k = 0; k < j; k++) {
                if(e_dist(p[k], c) >= cr + EPS) {
                    ret = circle(p[i], p[j], p[k]);
                    c = ret.P;
                    cr = ret.r;
                }
            }
        }
    }
    return ret;
}
/*Geometry Algorithm*/
double DP[110][110];
double minCostPolygonTriangulation(polygon& A) {
    if(A.P.size() < 3)
        return 0;
    FOR(i, A.P.size()) {
        for(int j = 0, k = i; k < A.P.size(); j++, k++) {
            if(k < j + 2)
                DP[j][k] = 0.0;
            else {
                DP[j][k] = INF;
                REP(l, j + 1, k - 1) {
                    double cost = e_dist(A.P[j], A.P[k]) + e_dist(A.P[k], A.P[l]) + e_dist(A.P[l], A.P[j]);
                    DP[j][k] = min(DP[j][k], DP[j][l] + DP[l][k] + cost);
                }
            }
        }
    }
    return DP[0][A.P.size() - 1];
}

```

```

    for(j = 0; j < i; j++) {
        if(e_dist(p[j], c) >= cr + EPS) {
            c = mid(p[i], p[j]);
            cr = e_dist(p[i], c);
            ret = circle(c, cr);
            for(k = 0; k < j; k++) {
                if(e_dist(p[k], c) >= cr + EPS) {
                    ret = circle(p[i], p[j], p[k]);
                    c = ret.P;
                    cr = ret.r;
                }
            }
        }
    }
    return ret;
}
/*Geometry Algorithm*/
double DP[110][110];
double minCostPolygonTriangulation(polygon& A) {
    if(A.P.size() < 3)
        return 0;
    FOR(i, A.P.size()) {
        for(int j = 0, k = i; k < A.P.size(); j++, k++) {
            if(k < j + 2)
                DP[j][k] = 0.0;
            else {
                DP[j][k] = INF;
                REP(l, j + 1, k - 1) {
                    double cost = e_dist(A.P[j], A.P[k]) + e_dist(A.P[k], A.P[l]) + e_dist(A.P[l], A.P[j]);
                    DP[j][k] = min(DP[j][k], DP[j][l] + DP[l][k] + cost);
                }
            }
        }
    }
    return DP[0][A.P.size() - 1];
}

```

4.2 Convex Hull

```

typedef double TD; // for precision shifts
namespace GEOM {
    typedef pair<TD, TD> Pt; // vector and points
    const TD EPS = 1e-9;
    const TD maxD = 1e9;
    TD cross(Pt a, Pt b, Pt c) { // right hand rule
        TD v1 = a.first - c.first; // (a-c) X (b-c)
        TD v2 = a.second - c.second;
        TD u1 = b.first - c.first;
        TD u2 = b.second - c.second;
        return v1 * u2 - v2 * u1;
    }
    TD cross(Pt a, Pt b) { // a X b
        return a.first * b.second - a.second * b.first;
    }
    TD dot(Pt a, Pt b, Pt c) { // (a-c) . (b-c)
        TD v1 = a.first - c.first;
        TD v2 = a.second - c.second;
        TD u1 = b.first - c.first;
        TD u2 = b.second - c.second;
        return v1 * u1 + v2 * u2;
    }
    TD dot(Pt a, Pt b) { // a . b
        return a.first * b.first + a.second * b.second;
    }
    TD dist(Pt a, Pt b) {
        return sqrt((a.first - b.first) * (a.first - b.first) +

```

```

        (a.second - b.second) * (a.second - b.second));
    }
    TD shoelaceX2(vector<Pt>& convHull) {
        TD ret = 0;
        for(int i = 0, n = convHull.size(); i < n; i++)
            ret += cross(convHull[i], convHull[(i + 1) % n]);
        return ret;
    }
    vector<Pt> createConvexHull(vector<Pt>& points) {
        sort(points.begin(), points.end());
        vector<Pt> ret;
        for(int i = 0; i < points.size(); i++) {
            while(ret.size() > 1 &&
                cross(points[i], ret[ret.size() - 1], ret[ret.size() - 2]) < -EPS)
                ret.pop_back();
            ret.push_back(points[i]);
        }
        for(int i = points.size() - 2, sz = ret.size(); i >= 0; i--) {
            while(ret.size() > sz &&
                cross(points[i], ret[ret.size() - 1], ret[ret.size() - 2]) < -EPS)
                ret.pop_back();
            if(i == 0)
                break;
            ret.push_back(points[i]);
        }
        return ret;
    }
    bool isInside(Pt pv, vector<Pt>& x) { //using winding number
        int n = x.size(), wn = 0;
        x.push_back(x[0]);
        for(int i = 0; i < n; ++i) {
            if(((x[i + 1].first <= pv.first && x[i].first >= pv.first) ||
                (x[i + 1].first >= pv.first && x[i].first <= pv.first)) &&
                ((x[i + 1].second <= pv.second && x[i].second >= pv.second) ||
                (x[i + 1].second >= pv.second && x[i].second <= pv.second))) {
                if(cross(x[i], x[i + 1], pv) == 0) {
                    x.pop_back();
                    return true;
                }
            }
        }
        for(int i = 0; i < n; ++i) {
            if(x[i].second <= pv.second) {
                if(x[i + 1].second > pv.second && cross(x[i], x[i + 1], pv) > 0)
                    ++wn;
            } else if(x[i + 1].second <= pv.second && cross(x[i], x[i + 1], pv) < 0)
                --wn;
        }
        x.pop_back();
        return wn != 0;
    }
}
bool isInside(Pt pv, vector<Pt>& x) { //using winding number
    int n = x.size(), wn = 0;
    x.push_back(x[0]);
    for(int i = 0; i < n; ++i) {
        if(((x[i + 1].first <= pv.first && x[i].first >= pv.first) ||
            (x[i + 1].first >= pv.first && x[i].first <= pv.first)) &&
            ((x[i + 1].second <= pv.second && x[i].second >= pv.second) ||
            (x[i + 1].second >= pv.second && x[i].second <= pv.second))) {
            if(cross(x[i], x[i + 1], pv) == 0) {
                x.pop_back();
                return true;
            }
        }
    }
}
for(int i = 0; i < n; ++i) {
    if(x[i].second <= pv.second) {
        if(x[i + 1].second > pv.second && cross(x[i], x[i + 1], pv) > 0)
            ++wn;
    } else if(x[i + 1].second <= pv.second && cross(x[i], x[i + 1], pv) < 0)
        --wn;
}
x.pop_back();
return wn != 0;
}
}

```

```

    }
    x.pop_back();
    return wn != 0;
}
}

```

4.3 Closest Pair of Points

```

#define fi first
#define se second
typedef pair<int, int> pii;
struct Point {
    int x, y, id;
};
int compareX(const void* a, const void* b) {
    Point* p1 = (Point*)a, * p2 = (Point*)b;
    return (p1->x - p2->x);
}
int compareY(const void* a, const void* b) {
    Point* p1 = (Point*)a, * p2 = (Point*)b;
    return (p1->y - p2->y);
}
double dist(Point p1, Point p2) {
    return sqrt((double)(p1.x - p2.x) * (p1.x - p2.x) +
                (double)(p1.y - p2.y) * (p1.y - p2.y));
}
pair<pii, double> bruteForce(Point P[], int n) {
    double min = 1e8;
    pii ret = pii(-1, -1);
    for(int i = 0; i < n; ++i)
        for(int j = i + 1; j < n; ++j)
            if(dist(P[i], P[j]) < min) {
                ret = pii(P[i].id, P[j].id);
                min = dist(P[i], P[j]);
            }
    return pair<pii, double> (ret, min);
}
pair<pii, double> getmin(pair<pii, double> x, pair<pii, double> y) {
    if(x.fi.fi == -1 && x.fi.se == -1)
        return y;
    if(y.fi.fi == -1 && y.fi.se == -1)
        return x;
    return (x.se < y.se) ? x : y;
}
pair<pii, double> stripClosest(Point strip[], int size, double d) {
    double min = d;
    pii ret = pii(-1, -1);
    qsort(strip, size, sizeof(Point), compareY);
    for(int i = 0; i < size; ++i)
        for(int j = i + 1; j < size && (strip[j].y - strip[i].y) < min; ++j)
            if(dist(strip[i], strip[j]) < min) {
                ret = pii(strip[i].id, strip[j].id);
                min = dist(strip[i], strip[j]);
            }
    return pair<pii, double> (ret, min);
}
pair<pii, double> closestUtil(Point P[], int n) {
    if(n <= 3)
        return bruteForce(P, n);
    int mid = n / 2;
    Point midPoint = P[mid];
    pair<pii, double> dl = closestUtil(P, mid);
    pair<pii, double> dr = closestUtil(P + mid, n - mid);
    pair<pii, double> d = getmin(dl, dr);
    Point strip[n];
    int j = 0;
    for(int i = 0; i < n; ++i)
        if(abs(P[i].x - midPoint.x) < d.second)
            strip[j] = P[i], j++;
}

```

```

    return getmin(d, stripClosest(strip, j, d.second));
}
pair<pii, double> closest(Point P[], int n) {
    qsort(P, n, sizeof(Point), compareX);
    return closestUtil(P, n);
}
Point P[50005];
int main() {
    int n;
    scanf("%d", &n);
    for(int a = 0; a < n; a++) {
        scanf("%d%d", &P[a].x, &P[a].y);
        P[a].id = a;
    }
    pair<pii, double> hasil = closest(P, n);
    if(hasil.fi.fi > hasil.fi.se)
        swap(hasil.fi.fi, hasil.fi.se);
    printf("%d %d %.6lf\n", hasil.fi.fi, hasil.fi.se, hasil.se);
    return 0;
}

```

4.4 Smallest Enclosing Circle

```

// welzl's algo to find the 2d minimum enclosing circle of a set of points
// expected O(N)
// directions: remove duplicates and shuffle points, then call welzl(points)

struct Point {
    double x;
    double y;
};

struct Circle {
    double x, y, r;
    Circle() {}
    Circle(double _x, double _y, double _r): x(_x), y(_y), r(_r) {}
};

Circle trivial(const vector<Point>& r) {
    if(r.size() == 0)
        return Circle(0, 0, -1);
    else if(r.size() == 1)
        return Circle(r[0].x, r[0].y, 0);
    else if(r.size() == 2) {
        double cx = (r[0].x + r[1].x) / 2.0, cy = (r[0].y + r[1].y) / 2.0;
        double rad = hypot(r[0].x - r[1].x, r[0].y - r[1].y) / 2.0;
        return Circle(cx, cy, rad);
    } else {
        double x0 = r[0].x, x1 = r[1].x, x2 = r[2].x;
        double y0 = r[0].y, y1 = r[1].y, y2 = r[2].y;
        double d = (x0 - x2) * (y1 - y2) - (x1 - x2) * (y0 - y2);
        double cx = (((x0 - x2) * (x0 + x2) + (y0 - y2) * (y0 + y2)) / 2 *
            (y1 - y2) - ((x1 - x2) * (x1 + x2) + (y1 - y2) * (y1 + y2)) / 2 *
            (y0 - y2)) / d;
        double cy = (((x1 - x2) * (x1 + x2) + (y1 - y2) * (y1 + y2)) / 2 *
            (x0 - x2) - ((x0 - x2) * (x0 + x2) + (y0 - y2) * (y0 + y2)) / 2 *
            (x1 - x2)) / d;
        return Circle(cx, cy, hypot(x0 - cx, y0 - cy));
    }
}

// SHUFFLE THE POINTS FIRST!!!!!!
Circle welzl(const vector<Point>& p, int idx = 0, vector<Point> r = {}) {
    if(idx == (int) p.size() || r.size() == 3)
        return trivial(r);
    Circle d = welzl(p, idx + 1, r);
    if(hypot(p[idx].x - d.x, p[idx].y - d.y) > d.r) {
        r.push_back(p[idx]);
        d = welzl(p, idx + 1, r);
    }
}

```

```

    return d;
}

```

4.5 Centroid of Polygon

$$C_x = \frac{1}{6A} \sum_{i=0}^{n-1} (x_i + x_{i+1})(x_i y_{i+1} - x_{i+1} y_i)$$

$$C_y = \frac{1}{6A} \sum_{i=0}^{n-1} (y_i + y_{i+1})(x_i y_{i+1} - x_{i+1} y_i)$$

4.6 Pick Theorem

A: Area of a simply closed lattice polygon
 B: Number of lattice points on the edges
 I: Number of points in the interior
 $A = I + \frac{B}{2} - 1$

5 Graphs

5.1 Articulation Point and Bridge

```

// gr -> adj list
// vector vis, low -> initialize to -1
// int timer -> initialize to 0
void dfs(int pos, int dad = -1) {
    vis[pos] = low[pos] = timer++;
    int kids = 0;
    for(auto& i : gr[pos]) {
        if(i == dad)
            continue;
        if(vis[i] >= 0)
            low[pos] = min(low[pos], vis[i]);
        else {
            dfs(i, pos);
            low[pos] = min(low[pos], low[i]);
            if(low[i] > vis[pos])
                is_bridge(pos, i)
                if(low[i] >= vis[pos] && dad >= 0)
                    is_articulation_point(pos)
                    ++kids;
        }
    }
    if(dad == -1 && kids > 1)
        is_articulation_point(pos)
}

```

5.2 SCC and Strong Orientation

```

#define N 10020
vector<int> adj[N];
bool vis[N], ins[N];
int disc[N], low[N], gr[N];
stack<int> st;
int id, grid;
void scc(int cur, int par) {
    disc[cur] = low[cur] = ++id;
    vis[cur] = ins[cur] = 1;
    st.push(cur);
    for(int to : adj[cur]) {
        //if (to==par) continue; // ini untuk SO(scc undirected)
        if(!vis[to])
            scc(to, cur);
        if(ins[to])
            low[cur] = min(low[cur], low[to]);
    }
    if(low[cur] == disc[cur]) {
        grid++; // group id
        while(ins[cur]) {

```

```

    gr[st.tp] = grid;
    ins[st.tp] = 0;
    st.pop();
  }
}

```

5.3 Bipartite Graphs

|MCBM| = Maximum Cardinality Bipartite Matching

|Minimum Vertex Cover| = |MCBM|

|Maximum Independent Set| = |V| - |MCBM|

|Minimum Path Cover| = |V| - |MCBM|

In a bipartite graph, a set of vertices S is a vertex cover if and only if its complement $V-S$ is an independent set.

In a DAG, a set of vertices P is a path cover if and only if its complement $V-P$ is an independent set.

To find the Minimum Vertex Cover from the MCBM, find all vertices not matched on the left side, perform a DFS traversing

5.4 Dinic's Maximum Flow

```

// O(VE log(max_flow)) if scaling == 1
// O((V + E) sqrt(E)) if unit graph (turn scaling off)
// O((V + E) sqrt(V)) if bipartite matching (turn scaling off)
// indices are 0-based
template <typename T>
struct Dinic {

```

```

    struct Edge {
        int v, rev;
        T cap, flow;
    };

```

```

    int n;
    vector<vector<Edge>> g;
    vector<int> lvl, it;
    static constexpr T eps = (T)1e-9;
    bool scaling;

```

```

    Dinic(int n, bool scaling = true): n(n), g(n), lvl(n), it(n), scaling(scaling) {}

```

```

    void add_edge(int u, int v, T cap, bool directed = true) {
        Edge a{v, (int)g[v].size(), cap, 0};
        Edge b{u, (int)g[u].size(), directed ? 0 : cap, 0};
        g[u].push_back(a);
        g[v].push_back(b);
    }

```

```

    bool bfs(int s, int t, T lim) {
        fill(lvl.begin(), lvl.end(), -1);
        queue<int> q; q.push(s);
        lvl[s] = 0;
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (auto &e : g[u]) {
                if (lvl[e.v] == -1 && e.cap - e.flow >= lim + eps) {
                    lvl[e.v] = lvl[u] + 1;
                    q.push(e.v);
                }
            }
        }
        return lvl[t] != -1;
    }

```

```

    T dfs(int u, int t, T f) {
        if (u == t || f <= eps) return f;
        for (int &i = it[u]; i < (int)g[u].size(); i++) {
            Edge &e = g[u][i];
            if (lvl[e.v] == lvl[u] + 1 && e.cap - e.flow > eps) {

```

```

                T ret = dfs(e.v, t, min(f, e.cap - e.flow));
                if (ret > eps) {
                    e.flow += ret;
                    g[e.v][e.rev].flow -= ret;
                    return ret;
                }
            }
        }
        return 0;
    }

    T max_flow(int s, int t) {
        T flow = 0;
        T lim = scaling ? (T)(1 << 30) : 1;
        while (lim > eps) {
            while (bfs(s, t, lim)) {
                fill(it.begin(), it.end(), 0);
                while (true) {
                    T pushed = dfs(s, t, numeric_limits<T>::max());
                    if (pushed <= eps) break;
                    flow += pushed;
                }
            }
            lim /= 2;
        }
        return flow;
    }

    vector<int> min_cut(int s) {
        vector<int> vis(n, 0);
        queue<int> q; q.push(s); vis[s] = 1;
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (auto &e : g[u])
                if (!vis[e.v] && e.cap - e.flow > eps)
                    vis[e.v] = 1, q.push(e.v);
        }
        return vis; // vis[i] == 1 -> S-side
    }
};

```

5.5 MCMF Backtrack

```

// ---[ Min Cost Max Flow (Dijkstra + potentials, with path extraction) ]---
template<class FlowT = int, class CostT = long long>
struct MCMF { // In case CE, MCMF<ll,ll> mcmf;
    static constexpr CostT INF = numeric_limits<CostT>::max() / 4;
    static constexpr CostT EPS = (CostT)1e-9;

```

```

    struct E {
        int to, rev;
        FlowT cap, flow;
        CostT cost;
    };

```

```

    int n, s, t;
    vector<vector<E>> g;
    vector<CostT> dist, pot;
    vector<int> pv, pe;

```

```

    MCMF(int N, int S, int T): n(N), s(S), t(T),
        g(N), dist(N), pot(N), pv(N), pe(N) {}

```

```

    void addEdge(int u, int v, FlowT cap, CostT cost) {
        E a{v, (int)g[v].size(), cap, 0, cost};
        E b{u, (int)g[u].size(), 0, 0, -cost};
        g[u].push_back(a); g[v].push_back(b);
    }

```

```

    // optional: initialize potentials if negative costs exist

```

```

bool initPot() {
    fill(pot.begin(), pot.end(), INF);
    pot[s] = 0;
    for (int it = 0; it < n - 1; ++it) {
        bool any = false;
        for (int u = 0; u < n; ++u) if (pot[u] < INF/2) {
            for (auto &e : g[u]) if (e.cap > 0)
                if (pot[e.to] > pot[u] + e.cost) {
                    pot[e.to] = pot[u] + e.cost;
                    any = true;
                }
        }
        if (!any) break;
    }
    // check negative cycle
    for (int u = 0; u < n; ++u) if (pot[u] < INF/2)
        for (auto &e : g[u]) if (e.cap > 0)
            if (pot[e.to] > pot[u] + e.cost) return false;
    return true;
}

bool dijkstra(){
    fill(dist.begin(), dist.end(), INF);
    priority_queue<pair<CostT,int>, vector<pair<CostT,int>>, greater<>> pq;
    dist[s]=0; pq.push({0,s});
    while(!pq.empty()){
        auto [d,u]=pq.top(); pq.pop();
        if(d > dist[u] + EPS) continue;
        for(int i=0;i<(int)g[u].size();i++){
            E &e=g[u][i];
            if(e.cap > e.flow){
                CostT nd=d+e.cost+pot[u]-pot[e.to];
                if(nd+EPS < dist[e.to]){
                    dist[e.to]=nd; pv[e.to]=u; pe[e.to]=i;
                    pq.push({nd,e.to});
                }
            }
        }
    }
    return dist[t] < INF/2;
}

pair<FlowT,CostT> flow(){
    FlowT totF=0; CostT totC=0;
    while(dijkstra()){
        for(int i=0;i<n;i++) if(dist[i]<INF/2) pot[i]+=dist[i];
        FlowT add=numeric_limits<FlowT>::max();
        for(int v=t;v!=s;v=pv[v])
            add=min(add, g[pv[v]][pe[v]].cap - g[pv[v]][pe[v]].flow);
        for(int v=t;v!=s;v=pv[v]){
            E &e=g[pv[v]][pe[v]];
            e.flow+=add; g[v][e.rev].flow-=add;
            totC += (CostT)add * e.cost;
        }
        totF += add;
    }
    return {totF, totC};
}

// ---- backtrack flow decomposition ----
FlowT dfsPath(int u, FlowT f, vector<int> &path, vector<vector<E>> &res) const {
    if(u == t) return f;
    for(auto &e: res[u]) if(e.flow > EPS){
        FlowT pushed = dfsPath(e.to, min(f, e.flow), path, res);
        if(pushed > EPS){
            e.flow -= pushed;
            path.push_back(u);
            return pushed;
        }
    }
    return 0;
}

```

```

}

vector<pair<vector<int>,FlowT>> getPaths() const {
    auto res = g;
    vector<pair<vector<int>,FlowT>> out;
    while(true){
        vector<int> path;
        FlowT pushed = dfsPath(s, numeric_limits<FlowT>::max(), path, res);
        if(pushed <= EPS) break;
        reverse(path.begin(), path.end());
        path.push_back(t);
        out.push_back({path, pushed});
    }
    return out;
}
};

```

5.6 Flows with Demands

let S_0 be the source and T_0 be the original sink

1. add 2 additional nodes, call them S_1 and T_1
2. connect S_0 to nodes normally
3. connect nodes to T_0 normally
4. for each edge (U, V) , $\text{cap} = \text{original cap} - \text{demand}$
5. for each node N :
 1. add an edge (S_1, N) , $\text{cap} = \text{sum of inward demand to } N$
 2. add an edge (N, T_1) , $\text{cap} = \text{sum of outward demand from } N$
6. add an edge (T_0, S_0) , $\text{cap} = \text{INF}$
7. the above is not a typo!
8. run max flow normally
9. for each edge (S_1, V) and (U, T_1) , check if $\text{flow} == \text{cap}$

if step #9 fails, then it is not possible to satisfy the given demand

Mathematically, let $d(e)$ be the demand of edge e . Let V be the set of every vertex in the graph.

- $c'(S_1, v) = \sum_{u \in V} d(u, v)$ for each edge (s', v) .
- $c'(v, T_1) = \sum_{w \in V} d(v, w)$ for each edge (v, t') .
- $c'(u, v) = c(u, v) - d(u, v)$ for each edge (u, v) in the old network.
- $c'(T_0, S_0) = \infty$

5.7 Eulerian Path or Cycle

```

// finds a eulerian path / cycle
// visits each edge only once
// properties:
// - cycle: degrees are even
// - path: degrees are even OR degrees are even except for 2 vertices
// how to use: g = adjacency list g[n] = connected to n, undirected
// if there is a vertex u with an odd degree, call dfs(u)
// else call on any vertex
// ans = path result

```

```

vector<set<int>> g;
vector<int> ans;

```

```

void dfs(int u) {
    while(g[u].size()) {
        int v = *g[u].begin();
        g[u].erase(v);
        g[v].erase(u);
        dfs(v);
    }
    ans.push_back(u);
}

```

5.8 2-SAT

```
struct TwoSAT {
    int n;
    vector<vector<int>> g, gr;
    vector<int> comp, topological_order, answer;
    vector<bool> vis;

    TwoSAT() {}
    TwoSAT(int _n) :
        n(_n), g(2 * n), gr(2 * n), comp(2 * n), answer(2 * n), vis(2 * n) {}

    void add_edge(int u, int v) {
        g[u].push_back(v);
        gr[v].push_back(u);
    }

    // For the following three functions
    // int x, bool val: if 'val' is true, we take the variable to be x.
    // Otherwise we take it to be x's complement.

    // At least one of them is true
    void add_clause_or(int i, bool f, int j, bool p) {
        add_edge(i + (f ? n : 0), j + (p ? 0 : n));
        add_edge(j + (p ? n : 0), i + (f ? 0 : n));
    }

    // Only one of them is true
    void add_clause_xor(int i, bool f, int j, bool p) {
        add_clause_or(i, f, j, p);
        add_clause_or(i, !f, j, !p);
    }

    // Both of them have the same value
    void add_clause_and(int i, bool f, int j, bool p) {
        add_clause_xor(i, !f, j, p);
    }

    // Topological sort
    void dfs(int u) {
        vis[u] = true;
        for(const auto& v : g[u])
            if(!vis[v])
                dfs(v);
        topological_order.push_back(u);
    }

    // Extracting strongly connected components
    void scc(int u, int id) {
        vis[u] = true;
        comp[u] = id;
        for(const auto& v : gr[u])
            if(!vis[v])
                scc(v, id);
    }

    bool satisfiable() {
        fill(vis.begin(), vis.end(), false);
        for(int i = 0; i < 2 * n; i++)
            if(!vis[i])
                dfs(i);
        fill(vis.begin(), vis.end(), false);
        reverse(topological_order.begin(), topological_order.end());
        int id = 0;
        for(const auto& v : topological_order)
            if(!vis[v])
                scc(v, id++);
        // Constructing the answer
        for(int i = 0; i < n; i++) {
            if(comp[i] == comp[i + n])
                return false;
        }
    }
};
```

```
        answer[i] = (comp[i] > comp[i + n] ? 1 : 0);
    }
    return true;
}
};
```

6 Math

6.1 Extended Euclidean GCD

```
// computes x and y such that ax + by = gcd(a, b) in O(log (min(a, b)))
// returns {gcd(a, b), x, y}
tuple<int, int, int> gcd(int a, int b) {
    if(b == 0) return {a, 1, 0};
    auto [d, x1, y1] = gcd(b, a % b);
    return {d, y1, x1 - y1 * (a / b)};
}
```

6.2 Generalized CRT

```
template<typename T>
T extended_euclid(T a, T b, T& x, T& y) {
    if(b == 0) {
        x = 1;
        y = 0;
        return a;
    }
    T xx, yy, gcd;
    gcd = extended_euclid(b, a % b, xx, yy);
    x = yy;
    y = xx - (yy * (a / b));
    return gcd;
}

template<typename T>
T MOD(T a, T b) {
    return (a % b + b) % b;
}

// return x, lcm. x = a % n && x = b % m
template<typename T>
pair<T, T> CRT(T a, T n, T b, T m) {
    T _n, _m;
    T gcd = extended_euclid(n, m, _n, _m);
    if(n == m) {
        if(a == b)
            return pair<T, T>(a, n);
        else
            return pair<T, T>(-1, -1);
    } else if(abs(a - b) % gcd != 0)
        return pair<T, T>(-1, -1);
    else {
        T lcm = m * n / gcd;
        T x = MOD(a + MOD(n * MOD(_n * ((b - a) / gcd), m / gcd), lcm), lcm);
        return pair<T, T>(x, lcm);
    }
}
```

6.3 Linear Diophantine

```
//FOR SOLVING MINIMUM ABS(X) + ABS(Y)
ll x, y, newX, newY, target = 0;
ll extGcd(ll a, ll b) {
    if(b == 0) {
        x = 1, y = 0;
        return a;
    }
    ll ret = extGcd(b, a % b);
```

```

    newX = y;
    newY = x - y * (a / b);
    x = newX;
    y = newY;
    return ret;
}
ll fix(ll sol, ll rt) {
    ll ret = 0;
    //CASE SOLUTION(X/Y) < TARGET
    if(sol < target)
        ret = -floor(abs(sol + target) / (double)rt);
    //CASE SOLUTION(X/Y) > TARGET
    if(sol > target)
        ret = ceil(abs(sol - target) / (double)rt);
    return ret;
}
ll work(ll a, ll b, ll c) {
    ll gcd = extGcd(a, b);
    ll solX = x * (c / gcd);
    ll solY = y * (c / gcd);
    a /= gcd;
    b /= gcd;
    ll fi = abs(fix(solX, b));
    ll se = abs(fix(solY, a));
    ll lo = min(fi, se);
    ll hi = max(fi, se);
    ll ans = abs(solX) + abs(solY);
    for(ll i = lo; i <= hi; i++) {
        ans = min(ans, abs(solX + i * b) + abs(solY - i * a));
        ans = min(ans, abs(solX - i * b) + abs(solY + i * a));
    }
    return ans;
}

```

6.4 Modular Linear Equation

```

// finds all solutions to ax = b (mod n)
vi modular_linear_equation_solver(int a, int b, int n) {
    int x, y;
    vi ret;
    int g = extended_euclid(a, n, x, y);
    if(!(b % g)) {
        x = mod(x * (b / g), n);
        for(int i = 0; i < g; i++)
            ret.push_back(mod(x + i * (n / g), n));
    }
    return ret;
}

```

6.5 Miller-Rabin and Pollard's Rho

```

namespace MillerRabin {
    const vector<ll> primes = { // deterministic up to 2^64 - 1
        2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37
    };
    ll gcd(ll a, ll b) {
        return b ? gcd(b, a % b) : a;
    }
    ll powa(ll x, ll y, ll p) { // (x ^ y) % p
        if(!y)
            return 1;
        if(y & 1)
            return ((__int128) x * powa(x, y - 1, p)) % p;
        ll temp = powa(x, y >> 1, p);
        return ((__int128) temp * temp) % p;
    }
    bool miller_rabin(ll n, ll a, ll d, int s) {
        ll x = powa(a, d, n);
        if(x == 1 || x == n - 1)

```

```

        return 0;
        for(int i = 0; i < s; ++i) {
            x = ((__int128) x * x) % n;
            if(x == n - 1)
                return 0;
        }
        return 1;
    }
    bool is_prime(ll x) { // use this
        if(x < 2)
            return 0;
        int r = 0;
        ll d = x - 1;
        while((d & 1) == 0) {
            d >>= 1;
            ++r;
        }
        for(auto& i : primes) {
            if(x == i)
                return 1;
            if(miller_rabin(x, i, d, r))
                return 0;
        }
        return 1;
    }
}

namespace PollardRho {
    mt19937_64 generator(chrono::steady_clock::now()
        .time_since_epoch().count());
    uniform_int_distribution<ll> rand_ll(0, LLONG_MAX);
    ll f(ll x, ll b, ll n) { // (x^2 + b) % n
        return ((__int128) x * x) % n + b) % n;
    }
    ll rho(ll n) {
        if(n % 2 == 0)
            return 2;
        ll b = rand_ll(generator);
        ll x = rand_ll(generator);
        ll y = x;
        while(1) {
            x = f(x, b, n);
            y = f(f(y, b, n), b, n);
            ll d = MillerRabin::gcd(abs(x - y), n);
            if(d != 1)
                return d;
        }
    }
    void pollard_rho(ll n, vector<ll>& res) {
        if(n == 1)
            return;
        if(MillerRabin::is_prime(n)) {
            res.push_back(n);
            return;
        }
        ll d = rho(n);
        pollard_rho(d, res);
        pollard_rho(n / d, res);
    }
    vector<ll> factorize(ll n, bool sorted = 1) { // use this
        vector<ll> res;
        pollard_rho(n, res);
        if(sorted)
            sort(res.begin(), res.end());
        return res;
    }
}

```

6.6 Derangement


```

der[0] = 1;
der[1] = 0;
for(int i = 2; i <= 10; ++i)
    der[i] = (1ll)(i - 1) * (der[i - 1] + der[i - 2]);

```

6.7 Bernoulli Number

$$\sum_{k=1}^n k^m = \frac{1}{m+1} \sum_{i=0}^m \binom{m+1}{i} B_i^+ n^{m+1-i} = m! \sum_{i=0}^m \frac{B_i^+ n^{m+1-i}}{i!(m+1-i)!}$$

$$B_n^+ = 1 - \sum_{i=0}^{n-1} \binom{n}{i} \frac{B_i^+}{n-i+1}, \quad B_0^+ = 1$$

6.8 Forbenius Number

$(X * Y) - (X + Y)$ and total count is $(X - 1) * (Y - 1) / 2$

6.9 Stars and Bars with Upper Bound

$$P = (1 - X^{r_1+1}) \dots (1 - X^{r_n+1}) = \sum_i c_i X^{e_i}$$

$$Ans = \sum_i c_i \binom{N - e_i + n - 1}{n - 1}$$

6.10 Arithmetic Sequences

$$U_n = a + (n-1)a_1 + \frac{(n-1)(n-2)}{1 \times 2} a_2 + \dots + \frac{(n-1)(n-2)(n-3) \dots}{1 \times 2 \times 3 \times \dots} a_r$$

$$S_n = n \times a + \frac{n(n-1)}{1 \times 2} a_1 + \frac{n(n-1)(n-2)}{1 \times 2 \times 3} a_2 + \dots + \frac{n(n-1)(n-2)(n-3) \dots}{1 \times 2 \times 3 \dots} a_r$$

6.11 Basis Vector

```

int basis[d]; // basis[i] keeps the mask of the vector whose f value is i
int sz;       // Current size of the basis
void insertVector(int mask) {
    for(int i = 0; i < d; i++) {
        if((mask & 1 << i) == 0) {
            continue; // continue if i != f(mask)
        }
        if(!basis[i]) { // If there is no basis vector with the i'th bit set,
            // then insert this vector into the basis
            basis[i] = mask;
            ++sz;
            return;
        }
        mask ^= basis[i]; // Otherwise subtract the basis vector from this
        // vector
    }
}

```

7 Strings

7.1 Aho-Corasick

```

const int K = 26;
struct Vertex {
    int next[K];
    bool leaf = 0;
    int p = -1, ans = 0;
    char pch;
    int link = -1,mlink = -1;
    //magic link, is the link to find the nearest leaf
    int go[K];
    Vertex(int p = -1, char ch = '$') : p(p), pch(ch) {
        fill(begin(next), end(next), -1);
        fill(begin(go), end(go), -1);
    }
}

```

```

};
vector<Vertex> t;
int add_string(string const& s) {
    int v = 0;
    for(char ch : s) {
        int c = ch - 'a';
        if(t[v].next[c] == -1) {
            t[v].next[c] = t.size();
            t.emplace_back(v, ch);
        }
        v = t[v].next[c];
    }
    t[v].leaf = 1;
    return v;
}
int go(int v, char ch);
int get_link(int v) {
    if(t[v].link == -1) {
        if(v == 0 || t[v].p == 0)
            t[v].link = 0;
        else
            t[v].link = go(get_link(t[v].p), t[v].pch);
    }
    return t[v].link;
}
int get_mlink(int v) {
    if(t[v].mlink == -1) {
        if(v == 0 || t[v].p == 0)
            t[v].mlink = 0;
        else {
            t[v].mlink = go(get_link(t[v].p), t[v].pch);
            if(t[v].mlink && !t[t[v].mlink].leaf) {
                if(t[t[v].mlink].mlink == -1)
                    get_mlink(t[t[v].mlink]);
                t[v].mlink = t[t[v].mlink].mlink;
            }
        }
    }
    return t[v].mlink;
}
int go(int v, char ch) {
    int c = ch - 'a';
    if(t[v].go[c] == -1) {
        if(t[v].next[c] != -1)
            t[v].go[c] = t[v].next[c];
        else
            t[v].go[c] = v == 0 ? 0 : go(get_link(v), ch);
    }
    return t[v].go[c];
}
//t.pb(Vertex());

```

7.2 KMP

```

auto get_kmp = [&](string S, string T) -> vector<int> {
    // S is the text, T is the pattern // ababa aba -> expected: {1 2 3 2 3}
    int N = S.size(), M = T.size();
    vector<int> lps(M), kmp(N);
    for(int i = 1, j = 0; i < M; i++) {
        if(T[i] == T[j])
            lps[i++] = ++j;
        else {
            if(j)
                j = lps[j - 1];
            else
                lps[i++] = 0;
        }
    }
    for(int i = 0, j = 0; i < N; i++) {
        if(S[i] == T[j]) {

```

```
    kmp[i++] = ++j;
    if(j == M)
        j = lps[j - 1];
} else {
    if(j)
        j = lps[j - 1];
    else
        kmp[i++] = 0;
}
}
return kmp;
};
```

7.3 Z Function

```
auto z_function(string s) {
    int n = s.size();
    vector<int> z(n);
    int l = 0, r = 0;
    for(int i = 1; i < n; i++) {
        if(i < r) {
            z[i] = min(r - i, z[i - l]);
        }
        while(i + z[i] < n && s[z[i]] == s[i + z[i]]) {
            z[i]++;
        }
        if(i + z[i] > r) {
            l = i;
            r = i + z[i];
        }
    }
    return z;
}
```

8 OEIS

8.1 A000108 (Catalan)

Catalan numbers
 $f(n) = nCk(2n,n) / (n+1) = nCk(2n,n) - nCk(2n,n+1) = f(n-1) * 2*(2*n-1) / (n+1)$
1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, 208012, 742900,
2674440, 9694845, 35357670, 129644790, 477638700, 1767263190, 6564120420,
24466267020, 91482563640, 343059613650, 1289904147324, 4861946401452,
18367353072152, 69533550916004, 263747951750360, 1002242216651368,
3814986502092304

8.2 A018819

Binary partition function: number of partitions of n into powers of 2
 $f(2m+1) = f(2m); f(2m) = f(2m-1) + f(m)$
1, 1, 2, 2, 4, 4, 6, 6, 10, 10, 14, 14, 20, 20, 26, 26, 36, 36, 46, 46, 60,
60, 74, 74, 94, 94, 114, 114, 140, 140, 166, 166, 202, 202, 238, 238, 284,
284, 330, 330, 390, 390, 450, 450, 524, 524, 598, 598, 692, 692, 786, 786,
900, 900, 1014, 1014, 1154, 1154, 1294, 1294

8.3 A092098

3-Portolan numbers: number of regions formed by n-secting the angles of an equilateral triangle.
long long solve(long long n) {
 long long res = (n % 2 == 1 ? 3*n*n - 3*n + 1 : 3*n*n - 6*n + 6);
 const int bats = n/2 - 1;
 for (long long i=1; i<=bats; i++) for (long long j=1; j<=bats; j++) {
 long long num = i * (n-j) * n;
 long long denum = (n-i) * j + i * (n-j);
 res -= 6 * (num % denum == 0 && num / denum <= bats);
 }
}

```
    } return res;
}
```

1, 6, 19, 30, 61, 78, 127, 150, 217, 246, 331, 366, 469, 510, 625, 678, 817,
870, 1027, 1080, 1261, 1326, 1519, 1566, 1801, 1878, 2107, 2190, 2437, 2520,
2791, 2886, 3169, 3270, 3559, 3678, 3997, 4110, 4447, 4548, 4921, 5034, 5419,
5550, 5899, 6078, 6487

8.4 A000127

Maximal number of regions obtained by joining n points around a circle by straight lines
 $f(n) = (n^4 - 6*n^3 + 23*n^2 - 18*n + 24) / 24$
1, 2, 4, 8, 16, 31, 57, 99, 163, 256, 386, 562, 794, 1093, 1471, 1941, 2517,
3214, 4048, 5036, 6196, 7547, 9109, 10903, 12951, 15276, 17902, 20854, 24158,
27841, 31931, 36457, 41449, 46938, 52956, 59536, 66712, 74519, 82993, 92171,
102091, 112792, 124314

8.5 A001534

Number of graphs with n nodes and n edges.
0, 0, 1, 2, 6, 21, 65, 221, 771, 2769, 10250, 39243, 154658, 628635, 2632420,
11353457, 50411413, 230341716, 1082481189, 5228952960, 25945377057,
132140242356, 690238318754